

# Discrete Mathematics I

## Logic

Juhász Zsófia  
jzsofia@inf.elte.hu

Department of Computer Algebra

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# Discrete mathematics

The subfields of Discrete mathematics include:

- Logic
- Set theory
- Combinatorics
- Graph theory
- Number theory
- Algebra
- Cryptography
- Algorithm theory
- Theory of computation
- Information theory
- Game theory
- Discrete geometry
- Operations research
- Probability theory

# What do we study in this course?

## Four main topics:

1. **Foundations:** logic, sets, relations



2. **Complex numbers**

$$i^2 = -1$$

3. **Combinatorics**



4. **Graphs**



# Some Logic

**Mathematical logic** is part of the foundations of mathematics. It

- studies and formalises the **rules of correct reasoning** and
- defines a **formal language** in which mathematical ideas can be expressed in a clear and unambiguous way.

We can talk about mathematical logic of different orders:

- 0<sup>th</sup> order logic (propositional logic)
- 1<sup>st</sup> order logic (predicate logic)
- 2<sup>nd</sup> order logic

⋮

We are going to look at elements of 0<sup>th</sup> and 1<sup>st</sup> order logic.

# Logical operations

In logic every **statement** (or **proposition**) has a **truth value** that is either **true** or **false**.

Propositions can be connected by **logical operations**:

## Logical operations

- **negation**, notation:  $\neg p$ .
- **and** (or **conjunction**), notation:  $p \wedge q$ .
- **or** (**inclusive or** or **disjunction**), notation:  $p \vee q$ .
- **if ... then ...** (or **implication** or **conditional**), notation:  $p \Rightarrow q$ .
- **... if and only if ...** (or **equivalence**, **biconditional**), notation:  $p \Leftrightarrow q$ .

Logical operations can be defined by their truth tables:

| $p$ | $\neg p$ |
|-----|----------|
| $T$ | $F$      |
| $F$ | $T$      |

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ | $p \Rightarrow q$ | $p \Leftrightarrow q$ |
|-----|-----|--------------|------------|-------------------|-----------------------|
| $T$ | $T$ | $T$          | $T$        | $T$               | $T$                   |
| $T$ | $F$ | $F$          | $T$        | $F$               | $F$                   |
| $F$ | $T$ | $F$          | $T$        | $T$               | $F$                   |
| $F$ | $F$ | $F$          | $F$        | $T$               | $T$                   |

# Logical operations: different kinds of 'or'

Just like in everyday language, in logic, different types of **or** are used:

## Different types of 'or'

- **inclusive or**:  $p \vee q$  is true if and only if *at least either of*  $p$  and  $q$  is true.  
e.g.: 'If you like jazz or rock, join our music club.'
- **exclusive or**:  $p \oplus q$  is true if and only if *exactly one of*  $p$  and  $q$  is true.  
e.g.: 'Here we need to turn left or right.'
- **conflicting or**:  $p || q$  is true if and only if *at most one of*  $p$  and  $q$  is true.  
e.g.: 'Drink or drive!'

| $p$ | $q$ | $p \vee q$ | $p \oplus q$ | $p    q$ |
|-----|-----|------------|--------------|----------|
| $T$ | $T$ | $T$        | $F$          | $F$      |
| $T$ | $F$ | $T$        | $T$          | $T$      |
| $F$ | $T$ | $T$        | $T$          | $T$      |
| $F$ | $F$ | $F$        | $F$          | $T$      |

# Logical operations: implication

In logic, implication ( $p \Rightarrow q$ ) does **not mean causality**: the truth value of  $p \Rightarrow q$  depends only on the truth values of  $p$  and  $q$ .

| $p$ | $q$ | $p \Rightarrow q$ |
|-----|-----|-------------------|
| $T$ | $T$ | $T$               |
| $T$ | $F$ | $F$               |
| $F$ | $T$ | $T$               |
| $F$ | $F$ | $T$               |

## Examples:

- $2 \cdot 2 = 4 \Rightarrow i^2 = -1$ .
- $2 \cdot 2 = 5 \Rightarrow i^2 = -2$
- $2 \cdot 2 = -3 \Rightarrow \text{Dogs are mammals.}$

A logical operator can be expressed in more than one ways:

$$(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$$

## Proposition (Properties of logical operations/Laws of 0<sup>th</sup> order logic)

For every proposition  $p, q$  and  $r$  the following statements are always true (i.e. these statements are tautologies):

**Properties of  $\vee$  and  $\wedge$ :**

1. idempotence:  $p \vee p \Leftrightarrow p$   $p \wedge p \Leftrightarrow p$
2. associativity:  $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$   $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$
3. commutativity:  $(p \vee q) \Leftrightarrow (q \vee p)$   $(p \wedge q) \Leftrightarrow (q \wedge p)$
4. absorption laws:  $(p \vee q) \wedge p \Leftrightarrow p$   $(p \wedge q) \vee p \Leftrightarrow p$
5. De Morgan's laws:  $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$   $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$
  
6. distributivity:  $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$   
 $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$

**Inference rules:**

7. law of contrapositive:  $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$
8. modus ponens:  $((p \Rightarrow q) \wedge p) \Rightarrow q$
9. syllogism:  $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
10.  $((p \Rightarrow q) \wedge (q \Rightarrow p)) \Leftrightarrow (p \Leftrightarrow q)$

# Proof example: associativity of $\vee$

## Proof (Example: associativity of $\vee$ )

2  $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$

| $p$ | $q$ | $r$ | $q \vee r$ | $p \vee (q \vee r)$ | $p \vee q$ | $(p \vee q) \vee r$ | $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$ |
|-----|-----|-----|------------|---------------------|------------|---------------------|-------------------------------------------------------|
| $T$ | $T$ | $T$ | $T$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $T$ | $T$ | $F$ | $T$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $T$ | $F$ | $T$ | $T$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $T$ | $F$ | $F$ | $F$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $F$ | $T$ | $T$ | $T$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $F$ | $T$ | $F$ | $T$        | $T$                 | $T$        | $T$                 | $T$                                                   |
| $F$ | $F$ | $T$ | $T$        | $T$                 | $F$        | $T$                 | $T$                                                   |
| $F$ | $F$ | $F$ | $F$        | $F$                 | $F$        | $F$                 | $T$                                                   |

**Homework:** Prove the other properties of logical operators. (At least a few, but any of them can be asked for at the exam.)

# First order logic: predicates

First order logic is an extension of zeroth order logic: here we can also use **predicates** and **quantifiers**.

## Examples: proposition vs. predicate

- 1 '2 is an even number' is a proposition. Its truth value is **true**.

Is ' $x$  is an even number' a proposition? Because  $x$  is a variable (unknown), we cannot say if it is true or false. So, it is not a proposition. It is called a **predicate**.

- 2 ' $6$  is less than  $5$ ' is a proposition. Its truth value is **false**.

' $x$  is less than  $y$ ' contains variables, hence we cannot tell if it is true or false. It is not a proposition, but a **predicate**.

**Informally:** A **predicate** is a property which an entity may have or a relationship which may exist between more than one entities.

# First order logic: predicates

## Definition (predicate)

An  **$n$ -ary predicate** is an  $n$ -variable function over a given set – the so called **universe (or domain) of discourse** – which returns either true or false for each possible choice of the input values.

(We will learn the formal definition of functions later in the course.)

## Examples

|              |                          |                   |                                                |
|--------------|--------------------------|-------------------|------------------------------------------------|
| $D()$        | The Sahara is a desert.  | nullary predicate | true                                           |
| $E(x)$       | $x$ is an even number.   | unary predicate   | $E(1)$ : false<br>$E(2)$ : true                |
| $F(x, y)$    | $x$ and $y$ are friends. | binary predicate  |                                                |
| $S(x, y, z)$ | $x + y = z$ .            | ternary predicate | $S(-1, 4, 9)$ : false<br>$S(1, -3, -2)$ : true |

# First order logic: quantifiers

## Quantifiers

- $\exists$  (existential quantifier): 'there exist(s)', 'there is/are'.
- $\forall$  (universal quantifier): '(for) every', '(for) all'.

## Examples

$B(x)$ : ' $x$  is a bird.'  $P(x)$ : ' $x$  is a parrot.'

1. 'Every parrot is a bird.'  
( 'For every  $x$  if  $x$  is a parrot then  $x$  is a bird.' )  
 $\forall x (P(x) \Rightarrow B(x))$
2. 'There exists a bird which is a parrot.'  
 $\exists x (B(x) \wedge P(x))$
3. 'Every bird is a parrot.'  
 $\forall x (B(x) \Rightarrow P(x))$
4. 'Not every bird is a parrot.'  
 $\neg (\forall x (B(x) \Rightarrow P(x)))$

# First order logic: formulas

Given a set of predicate symbols and a set of variable symbols we can define first order formulas as follows:

## Definition (first order formulas)

- Predicates are the simplest, so called elementary formulas.
- If  $\mathcal{A}$  and  $\mathcal{B}$  are formulas then  $(\neg \mathcal{A})$ ,  $(\mathcal{A} \vee \mathcal{B})$ ,  $(\mathcal{A} \wedge \mathcal{B})$ ,  $(\mathcal{A} \Rightarrow \mathcal{B})$  and  $(\mathcal{A} \Leftrightarrow \mathcal{B})$  are also formulas.
- If  $\mathcal{A}$  is a formula and  $x$  is a variable then  $\exists x \mathcal{A}$  and  $\forall x \mathcal{A}$  are also formulas.

E.g.:  $(\forall x(P(x) \Rightarrow B(x))) \wedge (\neg(\forall x(B(x) \Rightarrow P(x))))$

'Every parrot is a bird, but not every bird is a parrot.'

### Note:

- If it does not lead to ambiguity then brackets can be omitted.
- The outermost brackets in a formula are often omitted.

# Truth values of universally quantified formulas

## Definition (truth value of universally quantified formula)

Given a formula  $\mathcal{A}(x)$ , the universally quantified formula  $\forall x \mathcal{A}(x)$  is **true** if and only if  $\mathcal{A}(x)$  is true for every element of its universe of discourse; otherwise  $\forall x \mathcal{A}(x)$  is **false**.

**Example:** Predicates:  $E(x)$ : ' $x$  is even.' and  $Pr(x)$ : ' $x$  is prime.'

Decide about the truth value of the statement  $\forall x (E(x) \Rightarrow Pr(x))$  on the following universes of discourse:  $U_1 = \{1, 3, 5\}$  and  $U_2 = \{2, 4, 6\}$ .

①  $U_1 = \{1, 3, 5\}$

- $E(1) \Rightarrow Pr(1)$  true (since  $F \Rightarrow F$  is true)
- $E(3) \Rightarrow Pr(3)$  true (since  $F \Rightarrow T$  is true)
- $E(5) \Rightarrow Pr(5)$  true (since  $F \Rightarrow T$  is true)

Therefore  $\forall x (E(x) \Rightarrow Pr(x))$  is **true** on  $U_1$ .

②  $U_2 = \{2, 4, 6\}$

- $E(2) \Rightarrow Pr(2)$  true (since  $T \Rightarrow T$  is true)
- $E(4) \Rightarrow Pr(4)$  false (since  $T \Rightarrow F$  is false)

Therefore  $\forall x (E(x) \Rightarrow Pr(x))$  is **false** on  $U_2$ .